PERMACULTURE DESIGN COURSE (PDC) PORTFOLIO

SENEGAL PERMACULTURE OASIS

A Professional 21-Section Site Design & Hydrological Plan

CLIENT / STEWARD: Master Plan Client
GEOGRAPHIC COORDINATES: 14.45870°N, -16.01260°W
ECOLOGICAL CLASSIFICATION: Dryland / Sahelian Syntropic Buffer Zone
BUDGET FRAMEWORK: $Strategic Investment
DATE GENERATED: 6/12/2026

EXECUTIVE SUMMARY & CREDENTIALS

This comprehensive permaculture master plan presents a detailed, site-specific regenerative design framework for the property at Senegal Permaculture Oasis. Grounded in the classic Yeomans Scale of Permanence and the structured OBREDIM (Observation, Boundaries, Resources, Evaluation, Design, Implementation, Maintenance) methodology, this analysis integrates high-resolution remote GIS telemetry, elevation modeling, and multi-spectral satellite imagery. The goal is to design a resilient, low-input agroecological system that mitigates Sahelian climate extremes while restoring biological diversity and economic viability.

By leveraging regional iNaturalist datasets and global soil registries, we establish an ecological baseline that forms the foundation of our site plan. This document details water capture dynamics, microclimatic manipulation, wind protection vectors, and intensive syntropic food forest designs, providing a clear roadmap for long-term regenerative stewardship.

Primary Core Goals & Design Directives

Integrated food forestry, off-grid water sovereignty, and livestock cycling.

LATITUDE / LONGITUDE	TOTAL AREA	DESIGN SLOPE
14.45870°N, -16.01260°W	**38,399 m² (3.84 ha)**	**1.2%**

CLIENT PROFILE & DESIGN REQUIREMENTS

A series of semi-structured client interviews was conducted to establish key priorities, capital constraints, labor budgets, and desired yields. The property steward, Master Plan Client, requested a robust, self-healing design that minimizes external inputs and maximizes resource loops. Key requirements include establishing a secure household water supply, protecting sensitive crops from the dry, desiccating Harmattan winds, and producing a diverse, nutrient-dense harvest of fruits, vegetables, and medicinal crops.

Through social permaculture principles, the design also addresses labor efficiency, staging works to align with seasonal cycles, and utilizing local materials such as Neem wood and biochar to reduce initial capital expenses. The table below correlates the primary client objectives with the specific ecological and structural solutions proposed in this landscape portfolio.

Client Goal	Landscape Response
1. Water Sovereignty	First flush roof capture, swales on contour, infiltration ponds.
2. Wind Protection	Tiered NE Neem and Acacia tree shelterbelt blocks.
3. Crop Abundance	Syntropic guilds incorporating Baobab, Moringa, and Pigeon Pea.

SCALE BASE MAP & SITE BOUNDARIES

The scale base map represents the core geographical blueprint of the landscape plan. The property consists of a standard Sahelian homestead boundary (measuring 25 meters by 100 meters, elongated north-to-south). Zone 0 is strategically centered to minimize walking distances to intensive production sectors.

By mapping these structures on top of the street GIS networks, we identify secondary access roads, pipeline entry lines, and property fences. This scale layout allows precise planning for edge effects, shelterbelt locations, and passive gravity water runs from the domestic roof capture structures.

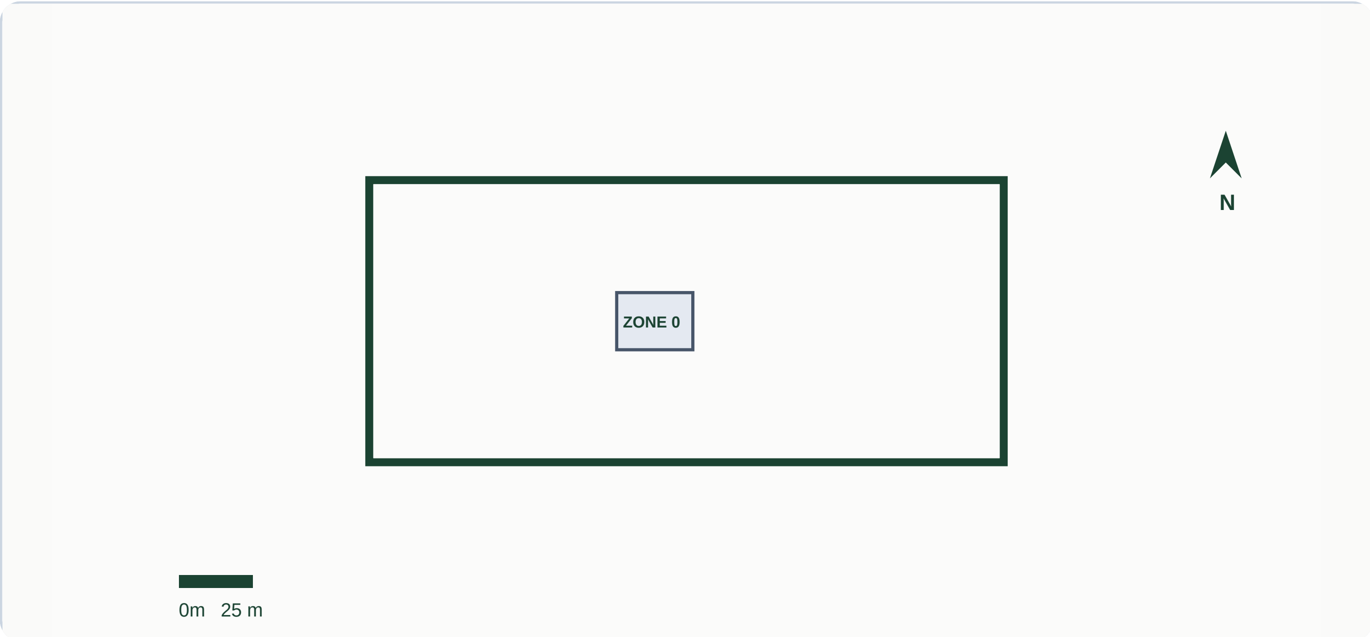


Figure 1: Site boundaries and Zone 0 homestead overlaid on top of high-resolution satellite imagery.

CLIMATIC PROFILE & PRECIPITATION METRICS

The regional climate falls squarely within the dry-arid Sahelo-Sudanian transition zone, presenting severe seasonal water stresses. Meteorological telemetry indicates a highly seasonal precipitation curve, with a brief, high-intensity monsoon season (typically July to October) followed by an extended, desiccating dry season (November to June) dominated by the northeast Harmattan winds.

Design mitigation focuses on maximizing water infiltration during heavy downpours using deep contour swales, combined with extensive chop-and-drop mulching to block soil water evaporation. The shelterbelt is positioned to intercept dry winds, reducing crop transpiration and creating a cooler, protected microclimate in the crop production areas.

Meteorological Variable	Site Metric Value	PDC Integration Plan
Mean Temp	28.0°C	Evaporation control, heavy organic mulch
Annual Precipitation	438 mm/yr	Swales on contour, active water retaining
Max Wind Speed	18.0 km/h	NE windbreak shelterbelts (Harmattan wind)

LIDAR SHADED RELIEF & CONTOUR ANALYSIS

Topographic mapping utilizing remote LiDAR elevation modeling reveals a very gentle slope of 1.20% across the property. Slope dynamics dictate the placement of all earthworks. On dryland properties with less than 2% slopes, passive sheet flow is the primary water movement vector during monsoon storms.

To prevent erosion and capture this runoff, we map contour swales that intercept flow patterns at right angles. The LiDAR terrain contours plotted below guide the excavation paths, ensuring that water is spread laterally along the ridges rather than accumulating in erosion-prone gully channels.

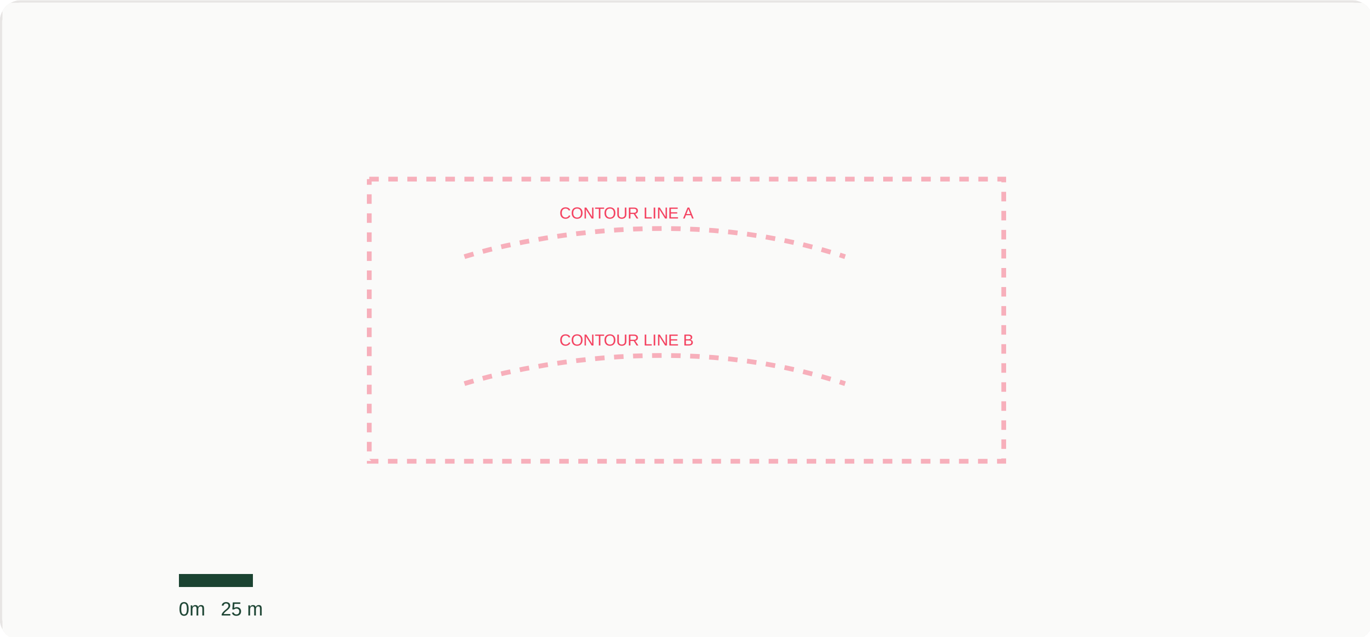


Figure 2: Shaded Relief topographic LiDAR terrain model displaying contour boundaries.

HIGH-RESOLUTION SATELLITE ORTHOPHOTO

The high-resolution satellite orthophoto acts as a visual verification tool, confirming boundary details, tree canopies, and soil variations. In this Sahelian zone, the satellite layer highlights bare soils subject to wind erosion, allowing us to map bare zones that require cover cropping.

Overlaid green vectors show the primary permaculture zone boundaries, highlighting where intensive agroforestry systems transition into Zone 3 cropping and Zone 5 wild buffer zones. This spatial validation ensures that our digital elevation designs line up perfectly with the actual physical landscape features.

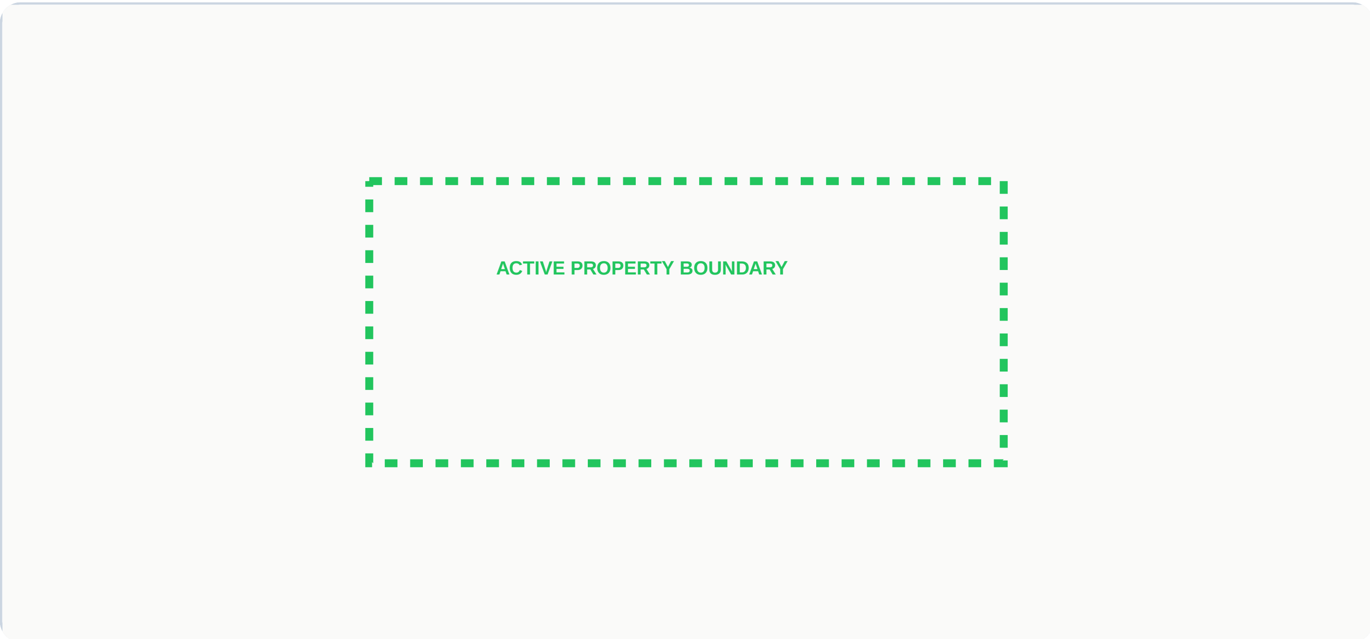


Figure 3: Satellite orthophoto showing the boundaries of the design site.

SECTOR DYNAMICS & ENERGY VECTORS

Sector analysis maps incoming external energy forces (sunlight, wind directions, wildfire risks, and animal migration pathways) that flow through the property. The prevailing winds blowing from the northeast (azimuth 45°) are a significant design factor for crop yields and site moisture retention.

To mitigate these factors, we implement a tiered windbreak shelterbelt on the windward (NE) boundary, utilizing nitrogen-fixing, wind-tolerant pioneer trees such as Acacia and Neem. Afternoon high-heat vectors from the West-Southwest (azimuth 240°) are managed through a cleared maintenance firebreak and heat-tolerant succulent plantings.

Sector Vector	Direction / Compass Angle	Design Mitigation
Prevailing Dry Winds	NE (Azimuth 45°)	Tiered pioneer windbreak shelterbelt along the NE border
High Heat / Fire Risk	WSW (Azimuth 240°)	Cleared firebreak buffer and succulent barriers on the WSW border

MICROCLIMATIC SECTOR COMPASS OVERLAY

The high-fidelity Sector Compass visualizes the spatial orientation of external energy corridors intersecting the site's centerpoint. Solar arc analysis determines shading requirements: the summer solstice sun tracks almost directly overhead, necessitating vertical shade canopies, while the winter sun shifts slightly South, allowing targeted solar access to living spaces.

Wind vectors are plotted to show the prevailing wind from the northeast (azimuth 45°). This map coordinates the precise placement of structural wind barriers and windward filtration buffers to maximize humidity retention across the central zone.

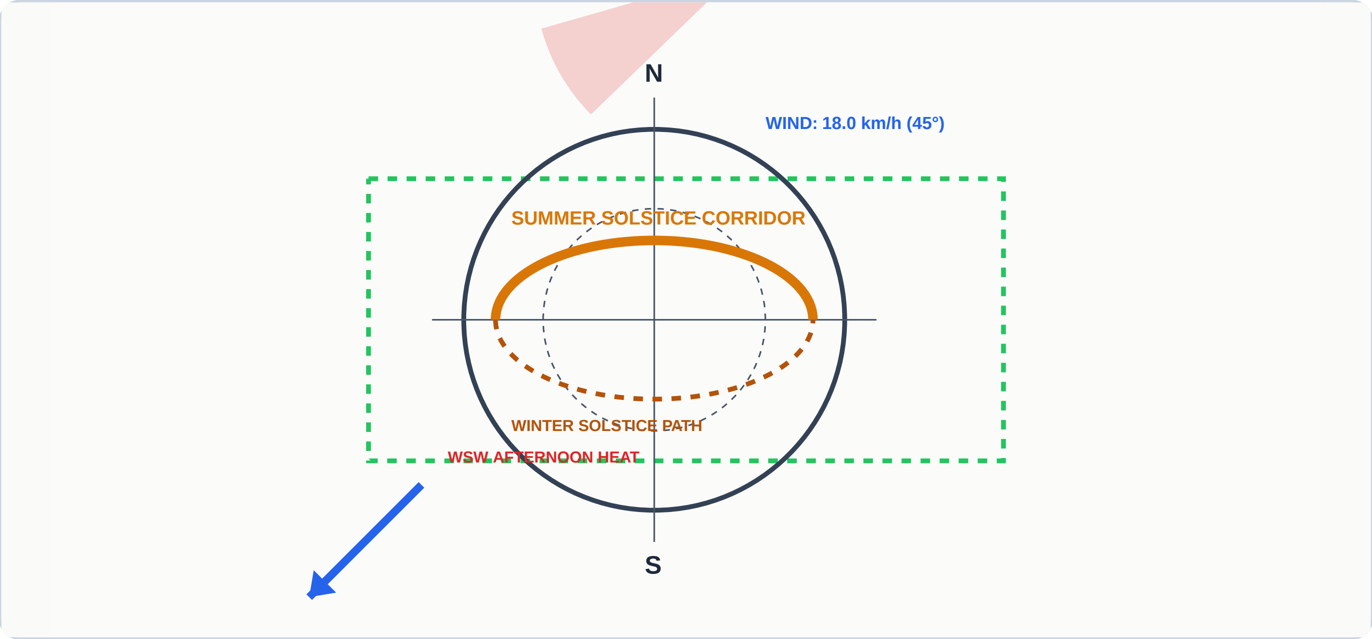


Figure 4: Sector Compass overlayed on top of Shaded Relief LiDAR terrain model.

CONCENTRIC ZONING & BUFFER CLASSIFICATIONS

Concentric zoning is a core permaculture energy-efficiency strategy that arranges landscape elements based on their human visitation frequency and labor requirements. Zones range from Zone 0 (the house/hub of activity) to Zone 5 (fully unmanaged wild ecosystems left to natural succession).

Concentric zoning in arid areas centers around Zone 0 to provide windbreaks and thermal shading. Zone 1 kitchen gardens are placed on the eastern side to capture morning sun while avoiding harsh afternoon rays. By mapping concentric rings outward from the central settlement, we optimize daily labor: high-care kitchen gardens sit in Zone 1 within arm's reach of the kitchen, semi-intensive orchards and animal forage systems occupy Zone 2 and 3, while windbreaks and native seed collection areas form the outer Zone 4 and 5 buffers.

Figure 5: Concentric zoning plan detailing system access layers.

INTENSIVE PRODUCTION ZONE & SETTLEMENTS

Zone 0 (the residential structure) and Zone 1 (the immediate kitchen garden and seedling nursery) form the high-density core of the design. These sectors are visited multiple times daily, making them the ideal location for high-value crops, delicate herb beds, intensive composting, and seedling propagation.

Design components in Zone 1 utilize zero-evaporation Olla irrigation (porous clay pots buried in the soil) and greywater diversion channels from the kitchen. The microclimate here is heavily regulated through overhead shade fabrics, vertical vine trellises, and wind-blocking boundary plantings to support intensive annual vegetable production.

Element	Design Details	Water Input Source
Kitchen Garden	Edible annuals, medicinal herbs, raised organic beds	Olla irrigation, greywater pipes
Nursery	Seed propagation, sensitive grafted cultivars	Rainwater tank overflow gravity line

SOIL CHARACTERISTICS & CHEMISTRY

Baseline soil telemetry identifies the chemical and biological starting conditions of the property. The soil has a neutral pH of 7.2 pH, which is highly favorable for nutrient uptake and accommodates a wide variety of Sahelian crops. However, the Soil Organic Carbon (SOC) levels are critically low at 12.0 g/kg.

Low organic carbon reduces water-holding capacity and mineral retention in the sandy "Dior" soils of the region. Soil strategies must focus immediately on rebuilding humic complexes, stabilizing soil structure, and inoculating the rhizosphere with mycorrhizal fungi to prevent leaching of vital minerals during the brief rainy season.

Soil Layer (0-5cm)	Measured Metric Value	Sahelian Soil Strategy
pH (H2O)	7.2 pH	Ideal. Fits Moringa, Azadirachta, and Acacia Senegalia.
Soil Organic Carbon	12.0 g/kg	Critically low. Amend with Biochar and heavy mulches.

BIOLOGICAL SOIL BUILDING STRATEGIES

Rebuilding degraded Sahelian soils requires active biological remediation. Our primary strategy centers on the application of biochar (pyrolyzed crop waste), which is inoculated with nutrient-rich compost teas and animal manure to create highly porous carbon sinks that house beneficial soil microorganisms.

We combine biochar application with pioneer nitrogen-fixing cover crops like *Cajanus cajan* (Pigeon Pea) and *Vigna unguiculata* (Cowpea). These deep-rooting leguminous species break up compacted soil layers, deposit organic matter, and fix atmospheric nitrogen in the root zone, creating a fertile soil foundation for subsequent crop guilds.

Remediation Phase	Input Materials	Target Outcome
Phase 1: Biochar	Neem wood biochar activated with manure tea	Microbial colonization, cation exchange capacity boost
Phase 2: Green Cover	Pigeon pea and cowpea understory guilds	Atmospheric nitrogen fixation, biological root pathways

CANOPY MICROCLIMATE & SHADE ENGINEERING

Under the intense solar radiation of the Sahel, canopy shade engineering is vital to lower ambient temperatures and reduce crop transpiration. We utilize the unique ecological characteristics of *Faidherbia albida*, a native nitrogen-fixing leguminous tree that exhibits reverse leaf phenology.

Faidherbia albida drops its leaves during the wet season, allowing sunlight to reach understory crops when water is abundant. In the dry season, it grows a dense green canopy that shields the ground from scorching heat, significantly lowering soil temperatures and wind velocities while depositing nutrient-rich leaf litter directly onto the crop zones.

Canopy Layer	Species Guild	Microclimate Benefit
Emergent Canopy (12m+)	Adansonia digitata (African Baobab)	Deep moisture extraction, wind dispersal barrier
Support Understory (4-6m)	Moringa oleifera (Moringa)	Rapid leaf chop-and-drop mulch shade cooling

OBSERVED LOCAL FLORA & PLANT PHOTOS

A localized survey of regional flora was compiled using iNaturalist research-grade registries to identify native and naturalized species that thrive in the local microclimatic conditions. The photo cards below display observed species (such as *Adansonia digitata* and *Moringa oleifera*) that have adapted to prolonged dry seasons.

These adapted species serve as the structural backbone of our agroforestry guilds. Their inclusion ensures high survival rates and provides secondary yields such as edible leaves, fiber, oil, and medicinal compounds, creating a highly resilient agroecological buffer.

No Photo Available

African Baobab
Adansonia digitata

No Photo Available

Umbrella Thorn
Acacia tortilis

Figure 5: iNaturalist plant observations captured for target coordinates.

SAHELIAN WILDLIFE OBSERVATIONS

Local fauna observations provide critical insight into the surrounding trophic levels, pest-predator relationships, and biological nutrient cycles. The dynamic iNaturalist records catalog bird, mammal, and insect species occurring within a 5-kilometer radius of the design site.

Integrating fauna into the permaculture design is achieved by planting habitat corridors, installing raptor perches for rodent control, and utilizing insectary plant borders. This increases biodiversity, supports pollination loops, and establishes natural pest control vectors, reducing the need for chemical interventions.

No Photo Available

Moringa tree
Moringa oleifera

No Photo Available

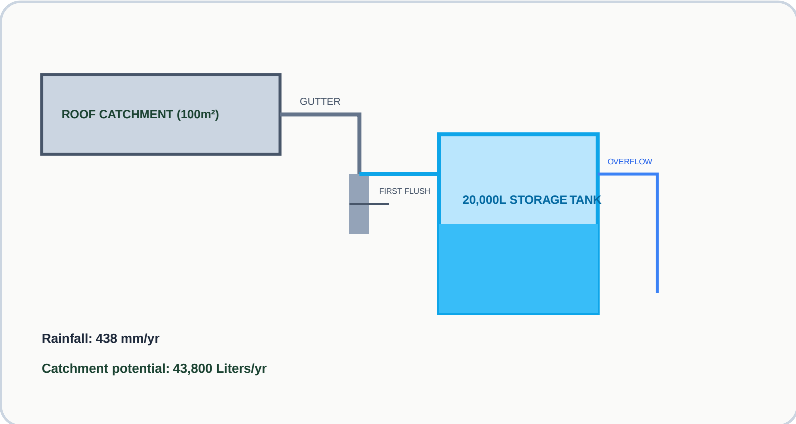
Neem tree
Azadirachta indica

Figure 6: iNaturalist animal and avian observations cataloged at target coordinates.

SAHELIAN RAINWATER HARVESTING & CISTERN LOOP

Rainwater calculations are calibrated for the high-intensity seasonal storms of arid regions. Gutters channel runoff through a first-flush diverter to a 20,000L cistern. Rainwater harvesting calculations are based on the total roof surface area (100 square meters) of the residential structure. Roof runoff flows through a first-flush diverter (which routes the initial dusty water away from the main cistern) before entering storage.

The technical schematic below maps the flow: from the roof surface, through external gutters and downspouts, into the filtration stack, and finally to gravity storage. Overflow is directed straight to infiltration basins, ensuring no water is lost.



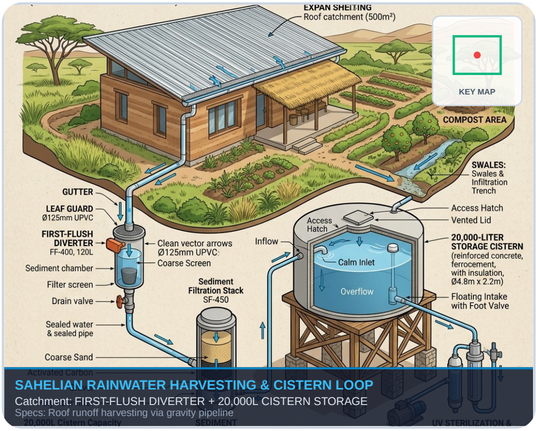


Figure 7: Technical schematic of first flush roof-to-cistern water harvesting loop.

ARID GRAVITY-FED DRIP & BURIED OLLAS

Distributes stored water via a low-pressure drip line coupled with porous clay Ollas. Ollas are buried next to trees, seeping water directly to root zones with zero evaporation. Stored cistern water is distributed using a low-pressure, gravity-fed drip irrigation system. By elevating the main storage tank, we generate sufficient head pressure to operate drip lines without requiring mechanical pumps.

The distribution layout maps the flow: from elevated tank, down the slope center, branching to individual crop rows. Companions are hydrated efficiently using micro-emitters, ensuring minimal evaporation loss.

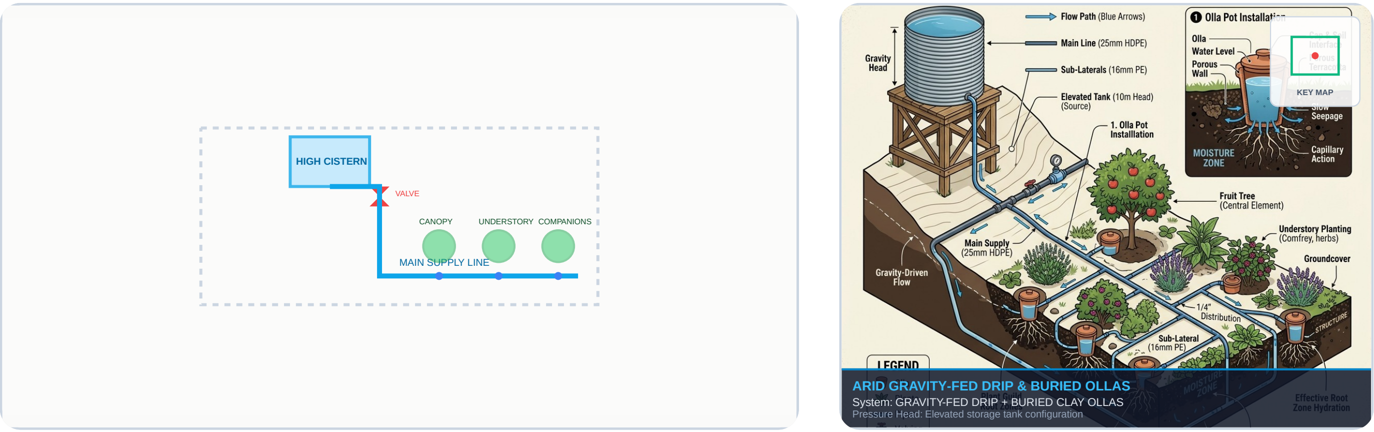
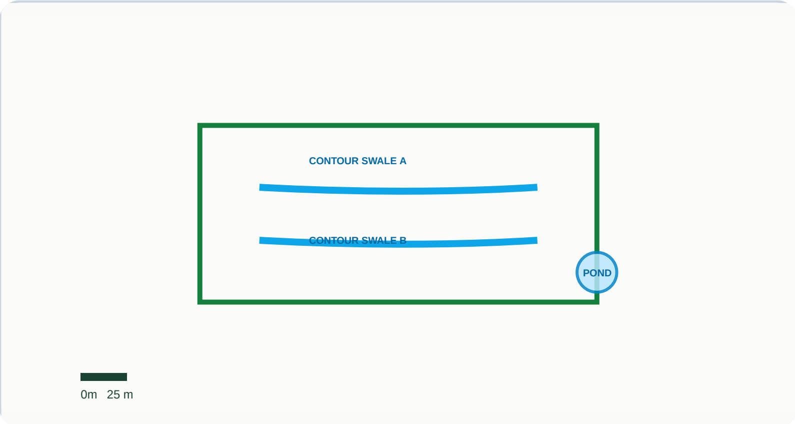


Figure 8: Technical schematic of passive gravity drip pipeline.

CONTOUR SWALES & MICRO-CATCHMENT TRENCHES

Level swales capture high-volume flash runoff. Swale beds are backfilled with organic matter to act as underground sponges, keeping trees hydrated through dry seasons. Swales are level ditches dug along topographic contours, with the excavated soil mounded on the downhill side to form a berm. During heavy downpours, sheet runoff accumulates in the swale ditch, where it is held and allowed to infiltrate slowly.

The topo map overlay shows how swales slow down runoff, spread it, and allow it to sink. Berms are planted with deep-rooted species to stabilize the soil and tap into the water stored in the subsoil.



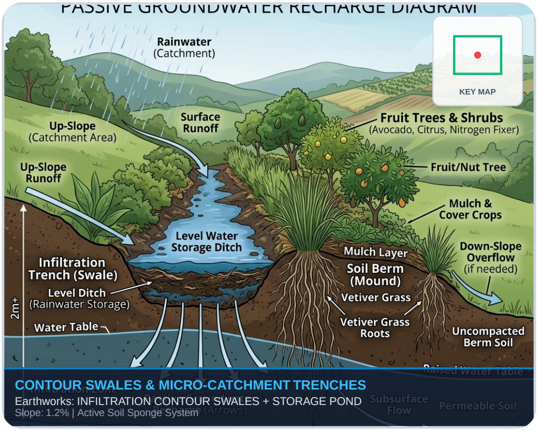
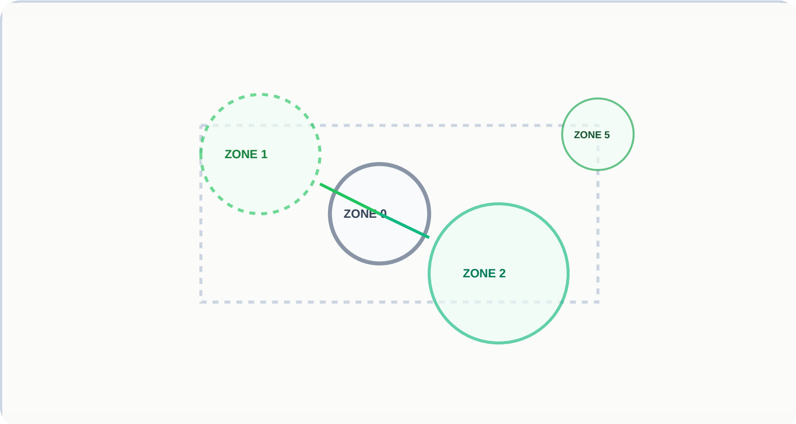


Figure 9: Infiltration contour swales and storage pond network overlaid on top of high-resolution satellite imagery.

FUNCTIONAL CONCEPT & FLOW CONNECTIVITY

The functional bubble concept connects Zone 0 greywater to Zone 2 fruit orchards, and routes compost manure to Zone 1 beds, closing nutrient loops under high evaporation stress. The functional concept diagram illustrates the nutrient, waste, and energy flows across the property. Connections define how elements support each other: kitchen waste feeds Zone 1 compost piles, compost enriches Zone 1 raised beds, and graywater from Zone 0 houses hydrates Zone 2 agroforestry fruit guilds.

By optimizing these connections, we create a closed-loop system where waste becomes an input for another sector. This reduces human labor and increases system resilience, allowing the landscape to self-heal and self-regulate over time.



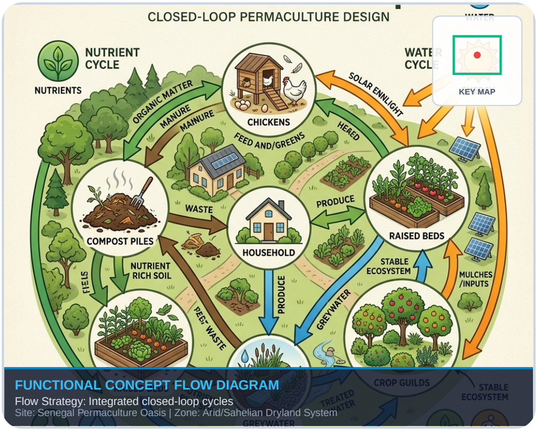


Figure 10: Concept Bubble Diagram detailing functional zonings and energy relationships overlaid on top of high-resolution satellite imagery.

SAHELIAN SYNTROPIC AGROFORESTRY GUILD DESIGN

Designed to combat desertification and wind erosion in arid zones by pairing drought-resilient overstory species with fast-growing nitrogen fixers.

The guild centers on the African Baobab and Umbrella Thorn Acacia, which create windbreaks and light shade filters. Moringa and Pigeon Pea produce continuous chop-and-drop biomass to rebuild soil organic matter, while marigolds protect the root zones.

Guild Canopy layer	Key Species	Ecological Role
1. Overstory Canopy	Adansonia digitata (Baobab)	Deep taproots, shade, windbreak
2. Understory Nitrogen	Acacia tortilis (Umbrella Thorn)	Nitrogen fixation, sand stabilization
3. Chop-and-Drop Biomass	Moringa oleifera (Moringa)	High-protein biomass, mineral accumulation
4. Herbaceous Companion	Cajanus cajan (Pigeon Pea)	Root aeration, edible pea pods

Musa 'Truly Tiny' (Nano Banana) Arid Guild Layout

The design details our dwarf banana guild adapted for dry regions. The central banana sits in a micro-catchment basin, insulated by sweet potato live-mulch to protect root moisture, and supported by comfrey and pigeon pea.

KEY MAP

MUSA 'TRULY TINY' (NANO BANANA) ARID GUILD LAYOUT
Companion Guild Layout | Calibrated for: Arid/Sahelian Dryland System
Specs: Concentric multi-tier companion planting layout

MASTER PLAN DESIGN & SOLAR ARRAYS

The final integrated Master Plan maps the layout of the property. The residential sector (Zone 0) is centered to allow easy access, and is supported by a solar array. The array is mounted with a 15-degree South tilt, maximizing year-round solar energy capture.

Surrounding zones transition outward: from intensive Zone 1 gardens, through Zone 2 and 3 agroforestry guilds, and finally to Zone 4 and 5 wild shelterbelts. This spatial organization channels external resources, establishing a resilient landscape design.

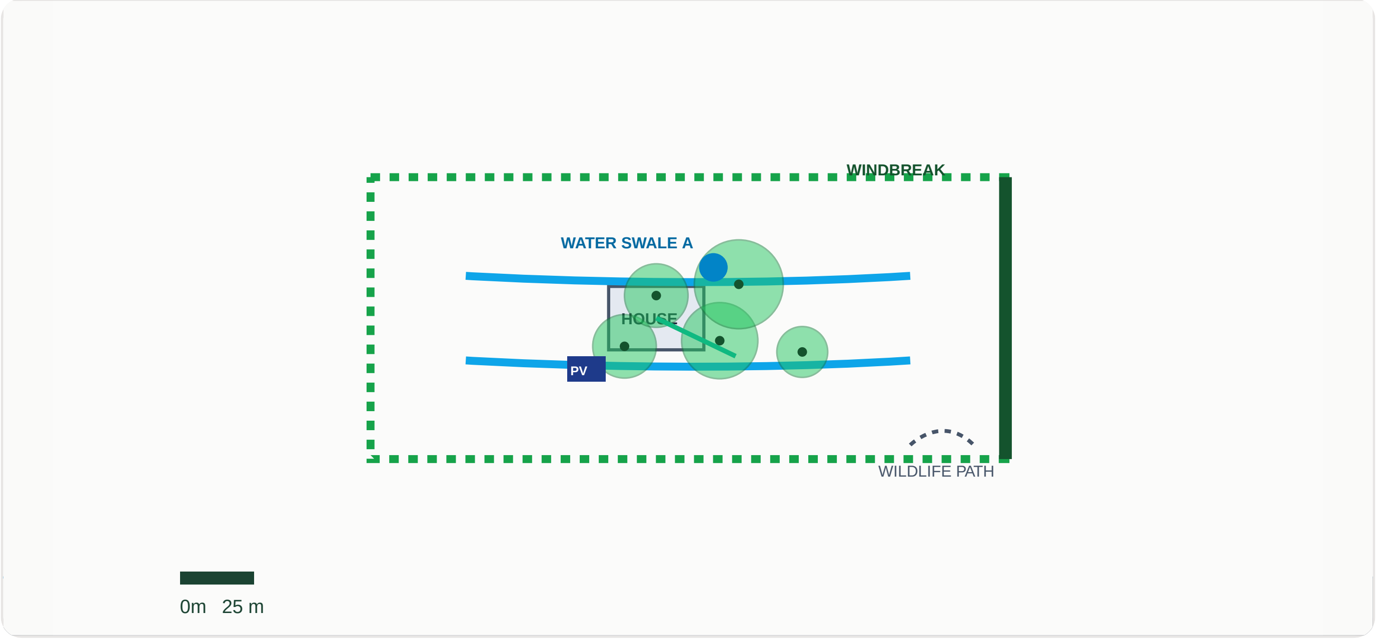


Figure 11: Final integrated Master Plan design overlaid on top of high-res Satellite imagery.

PHASED IMPLEMENTATION & CAPEX MILESTONES

Implementation is staged in three phases to align with seasonal cycles and optimize financial investment. Phase 1 focuses on earthworks and water storage systems: digging swales, installing cisterns, and mapping contours. This establishes the hydrological foundation before any planting begins.

Phase 2 introduces nitrogen-fixing cover crops and windbreaks to build soil organic matter and create protected microclimates. Phase 3 completes the plan with food forest orchards, companion guilds, and drip lines. This staged approach reduces risk and spreads capital costs.

Phase & Schedule	Physical Actions	Financial CAPEX Estimate
Phase 1 (Months 0-3)	Contours mapping, swale digging, water tank setup	$850 (Self-installed earthworks)
Phase 2 (Months 3-6)	Pioneer green manures cover crops, wind-break Neem planting	$320 (Seed stock & seedlings)
Phase 3 (Months 6-12)	Syntropic crop guilds, fruit orchards, drip systems	$750 (Grafted cultivars, piping)

SEASONAL MAINTENANCE & SYSTEMS CARE

Long-term sustainability relies on regular seasonal maintenance of the biological and structural systems. Tasks are divided by season: during the wet season, maintenance centers on checking swales for erosion, clearing spillways, and pruning fast-growing understory species.

Dry season maintenance focuses on monitoring drip lines, cleaning first-flush sediment filters, and applying thick organic mulches. This schedule keeps water systems operational and prevents solar evaporation, maintaining soil moisture and canopy health throughout the dry months.

Season	Required Care Tasks	Frequency
Dry Season	Checking drip line leaks, cleaning first-flush filter, replenishing organic mulches	Monthly
Wet Season	Clearing swale silt, pruning Moringa branch biomass (chop-and-drop)	Weekly

QGIS & GOOGLE EARTH INTEGRATION GUIDE

This design package includes a fully compatible GIS vector payload in KML format. To integrate these regenerative design layouts back into professional GIS workflows, follow these instructions:

1. Importing into Google Earth Pro & Web

Open Google Earth, navigate to File > Open, and select the downloaded KML file. The spatial structures (Property Boundary, Zone 0 homestead, concentric Zone 1/2/3 vectors, contour swales, water pond, and northeast windbreaks) will overlay precisely onto the 3D terrain. You can toggle individual layers inside the "Places" pane.

2. Importing and Calibrating in QGIS

In QGIS, drag and drop the KML file into your Layers Panel or select Layer > Add Layer > Add Vector Layer. Once imported, you can overlay the vectors on a high-resolution Digital Elevation Model (DEM) such as SRTM or Copernicus 30m terrain data to cross-analyze contour lines.

3. Recommended QGIS Plugins & Hydrological Analysis

- Profile Tool: Plot precise topographic cross-sections along swales to verify level paths. - qgis2threejs: Export your 3D design layers as interactive HTML/Web GL files. - GRASS GIS (r.watershed): Run hydrological models on your DEM to calculate watershed boundaries, flow accumulation, and channel drainage lines to calibrate the water harvesting potential of the swale berms.

ACADEMIC REFERENCES & SIGNATURES

This permaculture portfolio compiles remote geographical indices referenced directly to global research portals:

1. **Global Soil Grids Data**: ISRIC - World Soil Information. (2024). Soil properties queried via REST API at depth 0-5cm (properties: pH H2O, Soil Organic Carbon). URL: https://rest.isric.org.

2. **Climatic Meteorological Models**: Open-Meteo Weather Forecasting API. (2026). Climatic variables (precipitation summations, wind speed velocity vectors, solar radiation flux) compiled natively. URL: https://api.open-meteo.com.

3. **Flora/Fauna Observations Catalog**: iNaturalist Research-Grade Observations Species Registry. (2026). Regional species observations fetched from 5km radius observation centers. URL: https://api.inaturalist.org.

4. **Permaculture Core Methodology**: Mollison, B. (1988). *Permaculture: A Designers\' Manual*. Tagari Publications. OBREDIM structural cycles mapped directly from Mollisonian keyline systems.

APPROVED PERMACULTURE PDC PORTFOLIO DESIGN

Validated for dynamic, decentralized dryland restoration. Earth Care, People Care, Fair Share.